EE3032 Introduction to VLSI Design—Final

Fall 2023

Reconstruction from the two original printed pages; vector figures retain connectivity. The original point values sum to 115.

1. (30 points) Answer the following questions.

(a) List the components of dynamic power consumption of a dynamic logic gate. (6 points)

(b) What is physical design? (2 points)

(c) List two requirements for drawing the layout of a cell in the standard cell library. (4 points)

(d) List two design methods for avoiding glitch power of a CMOS circuit. (4 points)

(e) List two limitations of the dynamic logic. (4 points)

(f) What is the goal of the design rule? (2 points)

(g) Explain the function of filler cell, tap cell, decap cell, and endcap. (4 points)

(h) Illustrate two possible technologies which can be used to shorten manufacturing time such that the time to market can be shortened. (4 points)

2. (15 points) Fig. 1(a) shows a domino logic circuit and the corresponding clock timing diagram. Complete the following questions.

(a) If the output rise/fall times of the 3-input and 2-input AND gates are 3 ns/1 ns and 2 ns/1 ns, respectively, calculate the maximal operation frequency of the circuit. (5 points)

(b) The clock period of this circuit cannot be prolonged casually. Why? Modify the circuit such that its clock period can be prolonged casually. Draw the transistor schematic diagram. (5 points)

(c) Let the delay of each AND gate be 1 delay unit. If we apply the two input patterns (ABCD)=(0000,1110) in sequence, draw the output waveform with respect to input patterns. (5 points)

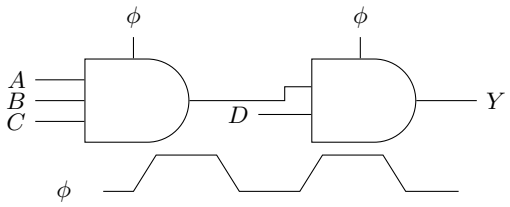


Figure 1(a): A domino logic circuit and its clock waveform.

3. (10 points) Fig. 1(b) shows a simple circuit with three gates. Complete the following questions.

(a) Can the circuit be operated correctly? Explain your answer. If your answer is "no", modify the circuit such that it can be operated correctly. (5 points)

(b) Assume that each transistor of the (modified) circuit has the following drain and gate capacitance: $C_d = 0.1\,\text{fF}$; $C_g = 1\,\text{fF}$; and the parasitic capacitance of the node x and y are $1.3\,\text{fF}$ and $1.1\,\text{fF}$, respectively. Let the switching voltage of all the gates in the (modified) circuit be $V_{DD}/2$. Can the circuit be operated correctly? Explain your answer. (5 points)

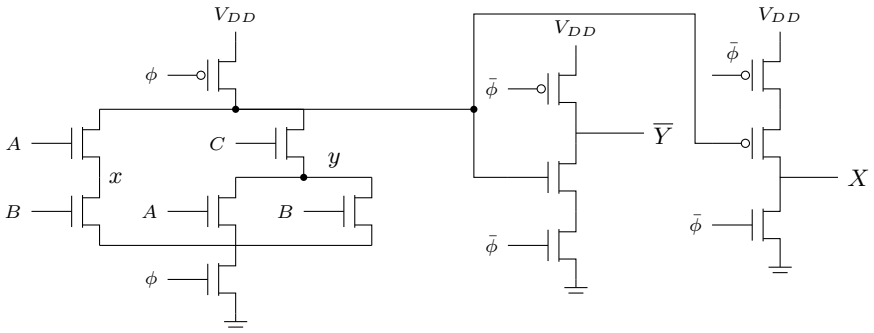


Figure 1(b): A simple circuit. The middle clock is transcribed as $\bar{\phi}$; its stroke touches the gate wire in the scan, so the solution also considers a ϕ reading.

4. (15 points) Design $f = \overline{x_1 x_2 + x_3}$ using (1) dynamic logic and (2) differential pass transistor logic; design

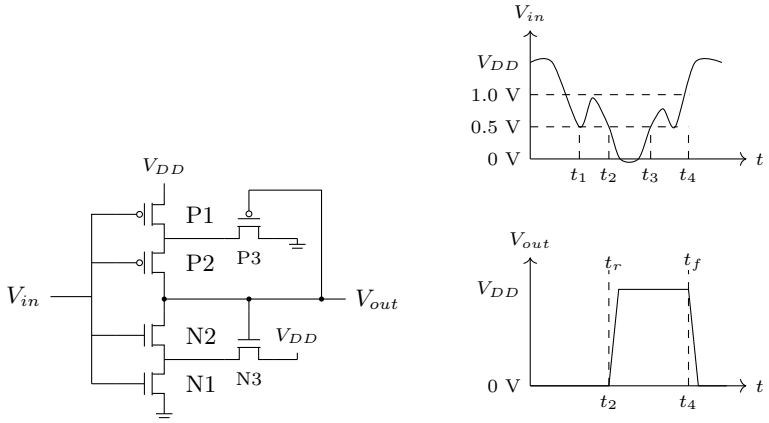
$$f = x_1 \oplus x_2 = \overline{x_1 x_2 + \overline{x_1 + x_2}}$$

using (3) NP domino logic.

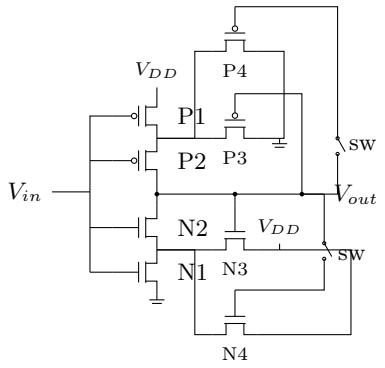
5. (15 points) Fig. 2(a) shows a Schmitt trigger circuit. If we apply an input signal as shown in the left figure to the Schmitt trigger circuit, the output signal will be as the right figure shows. Assume that $\mu_n = 2\mu_p$; the drain parasitic capacitance of all the transistors can be ignored; and the simple RC delay model is used to estimate the delay. Complete the following questions.

(a) Let $V_{tn} = |V_{tp}| = 0.2\,\text{V}$ and $V_{DD} = 1.5\,\text{V}$. If the size of N1 is 160 nm/40 nm, determine the sizes of the transistors N2, N3, P1, P2, and P3 such that $t_r = t_f = 3RC$, where the turn-on resistance and gate capacitance of N1 is R and C. All the transistors should have the same channel length. (5 points)

(b) Fig. 2(b) shows a Schmitt trigger circuit (STC) with controllable hysteresis voltage (V_H) by controlling the switches (sw). If the switches are turned on, is the noise tolerance capability of the STC increased or decreased? Is the short-circuit power of STC increased or decreased? Explain your answers. (10 points)



(a)



(b)

Figure 2: (a) A CMOS Schmitt trigger circuit (b) A STC with controllable hysteresis voltage.

6. (10 points) Draw the transistor-level schematic diagram of the symbolic layout shown in Fig. 3(a).

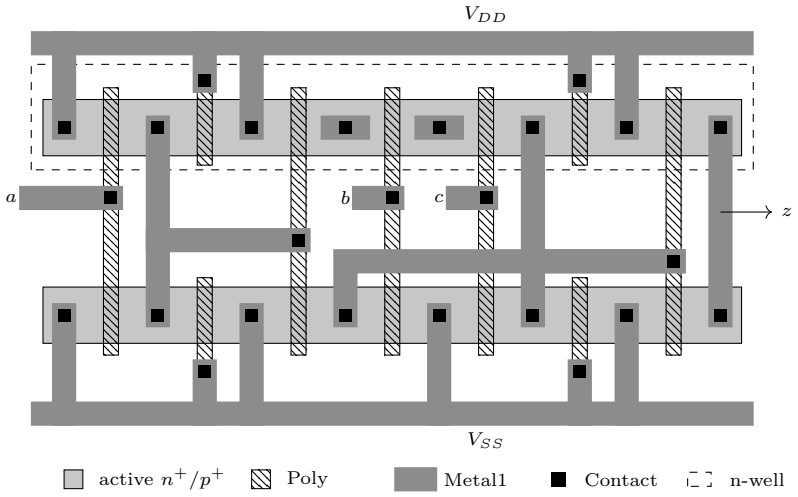


Figure 3(a): Symbolic layout. Normalized geometry preserves all contacts and all seven poly crossings per active row; scale is not specified in the original.

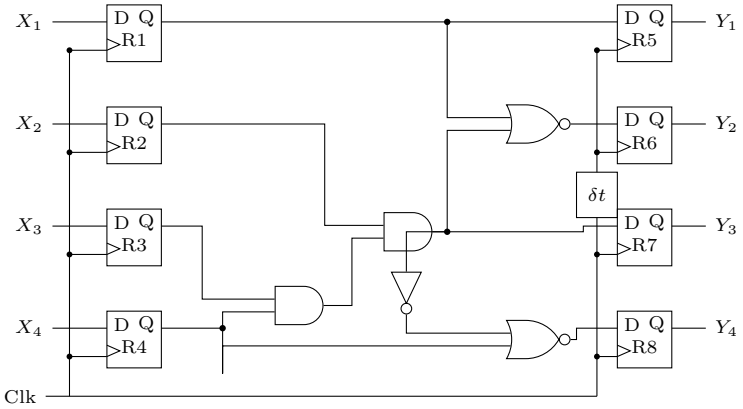


Figure 3(b): A sequential circuit. The inline capture-clock delay is below R6 and above R7.

7. (20 points) Fig. 3(b) shows a sequential circuit. Assume that the clock-to-Q propagation delay (t_{pcq}), the clock-to-Q contamination delay (t_{ccq}), the setup time (t_{setup}), and the hold time (t_{hold}) of the registers are 0.2 ns, 0.1 ns, 0.2 ns, and 0.2 ns, respectively. The delay of the AND gate, the NOR gate, and the inverter is 0.5 ns, 0.5 ns, and 0.2 ns, respectively. Also, the delay of interconnection can be ignored. Complete the following questions.

(a) Let $\delta t = 0$ ns. Can the circuit be operated correctly? If not, modify the circuit by adding a lockup latch such that it can operate correctly and draw the modified circuit. (5 points)

(b) Let $\delta t = 0$ ns. Determine the maximum operation frequency of the circuit or the modified circuit. (5 points)

(c) Let $\delta t = 0.3$ ns. Can the circuit be operated correctly? If yes, is the maximum operation frequency of the circuit changed? Explain your answers. (5 points)

(d) The registers typically are randomly distributed in a die. List two typical clock design methods for minimizing the clock skews among the registers. You should explain why the methods can minimize the clock skews. (5 points)